TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	81.8 ± 1.72
Correlation Stability	91.6 ± 0.80
Distribution Consistency	88.3 ± 2.00
Entropy Preservation	90.1 ± 1.14
Mean Z Score Drift	0.2 ± 0.02
Categorical Fidelity	85.1 ± 0.54
Privacy Risk	93.0 ± 0.00
Data Density	100.0 ± 0.01
Coverage Score	2.3 ± 0.78
Feature Accuracy	81.3 ± 2.05
Novelty Score	0.1 ± 0.00

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Dataset Audit: indian_liver_patient_dataset.csv

1. Dataset Portrait

Attribute	Value
Rows	583
Columns	11
Numeric features	10
Categorical features	1
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	40%
Generation % (*15)	100%
Join context (*16)	True

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	44.746	42.948	-1.798	4.0%	OK
TB	3.299	1.512	-1.787	54.2%	Drift
DB	1.486	0.622	-0.864	58.2%	Drift
Alkphos	290.576	225.869	-64.707	22.3%	Warn
Sgpt	80.714	38.212	-42.501	52.7%	Drift
Sgot	109.911	43.230	-66.681	60.7%	Drift
TP	6.483	6.748	+0.265	4.1%	OK
ALB	3.142	3.399	+0.257	8.2%	OK
A/G Ratio	0.947	0.936	-0.011	1.1%	OK
Selector	1.286	1.358	+0.072	5.6%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	81.8 ± 1.72
Correlation Stability (*2)	91.6 ± 0.80
Distribution Consistency (*3)	88.3 ± 2.00
Entropy Preservation (*4)	90.1 ± 1.14
Mean Z Score Drift (*5)	0.2 ± 0.02
Categorical Fidelity (*6)	85.1 ± 0.54
Privacy Risk (*7)	93.0 ± 0.00
Data Density (*8)	100.0 ± 0.01
Coverage Score (*9)	2.3 ± 0.78

Feature Accuracy (*10)	81.3 ± 2.05
Novelty Score (*11)	0.1 ± 0.00

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6743589743589744	0.5820512820512821	0.6880341880341882	0.013675213675213672	
F1	0.7821343746683586	0.6875778525200622	0.7962279847437549	0.01409361007539629	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7085470085470086	0.5034188034188034	0.6837606837606838	-0.02478632478632479	*
F1	0.8121854029423632	0.606881150591947	0.7934759035909326	-0.018709499351430715	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7059829059829059	0.5324786324786325	0.6931623931623931	-0.01282051282051282	
F1	0.8128002697439015	0.6283255582462217	0.7933189597901911	-0.0194813099537103	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376068	0.6914529914529914	0.7068376068376068	0	
F1	0.8279091624529537	0.8160754443804544	0.8279091624529537	0	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.558974358974359	0.570940170940171	0.5641025641025641	0.005128205128205125	
F1	0.5587432119380917	0.6528160141981537	0.5694202763846341	0.010677064446542494	

Model: MLP

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7	0.5641025641025641	0.6658119658119659	-0.03418803418803418	
F1	0.8035543512650885	0.6654417422915191	0.7628846200558866	-0.04066973120920179	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6632478632478633	0.5256410256410255	0.6230769230769231	-0.04017094017094015	
F1	0.7653544728427774	0.621693972494211	0.719008656541061	-0.04634581630171618	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6794871794871795	0.5017094017094017	0.6675213675213676	-0.01196581196581198	
F1	0.7869501953848472	0.6003820801206723	0.7717239932320645	-0.01522620215278264	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7025641025641025	0.5324786324786325	0.6598290598290599	-0.04273504273504271	
F1	0.7997903571549948	0.6363729724214344	0.7647365372622997	-0.03505381989269503	*

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6863247863247863	0.6068376068376069	0.6615384615384616	-0.0247863247863248	
F1	0.7971299319768493	0.7220393442269436	0.7805703407268142	-0.0165595912500351	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
2	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
3	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
4	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
5	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
6	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
7	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
8	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
9	17	Male	0.9	0.3	202	22	19	7.4	4.1	1.2
10	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
11	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
12	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
13	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
14	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
15	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
16	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
17	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
18	33	Male	1.6	0.5	165	15	23	7.3	3.5	0.92
19	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
20	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8

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B.1 Audit Trace: Fold 1 | Ctx: 40% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	49	Male	1.3	0.4	206	30	25	6.0	3.1	1.06
2	62	Male	1.8	0.9	224	69	155	8.6	4.0	0.8
3	48	Female	0.8	0.2	150	25	23	7.5	3.9	1.0
4	25	Male	0.8	0.1	130	23	42	8.0	4.0	1.0
5	45	Male	2.2	0.8	209	25	20	8.0	4.0	1.0
6	32	Male	12.7	8.4	190	28	47	5.4	2.6	0.9
7	34	Female	1.2	0.5	210	30	25	7.0	3.5	1.0
8	45	Male	2.5	1.0	300	50	60	8.0	4.0	1.2
9	29	Female	0.9	0.3	180	22	20	6.5	3.0	1.1
10	52	Male	3.0	1.5	250	40	45	7.5	3.8	1.3
11	39	Female	1.5	0.7	220	35	30	7.8	4.2	1.1
12	60	Male	0.8	0.2	190	25	28	6.0	2.9	0.9
13	33	Male	7.1	3.7	196	622	497	6.9	3.6	1.09
14	38	Male	2.2	1.0	310	119	42	7.9	4.1	1.0
15	64	Male	3.0	1.4	248	46	40	6.5	3.2	0.9
16	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8

B.2 Audit Trace: Fold 2 | Ctx: 40% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	60	Male	2.6	1.2	171	42	37	5.4	2.7	1.0
2	40	Male	0.7	0.1	202	37	29	5.0	2.6	1.0
3	38	Male	1.5	0.4	298	60	103	6.0	3.0	1.0
4	39	Male	6.6	3.0	215	190	950	4.0	1.7	0.7
5	45	Male	2.5	1.2	163	28	22	7.6	4.0	1.1
6	48	Male	5.8	2.5	802	133	88	6.0	2.8	0.8
7	29	Female	1.2	0.4	180	25	30	6.5	3.2	1.1
8	54	Male	3.5	1.5	250	45	60	7.0	3.8	0.9
9	38	Female	0.9	0.3	200	30	25	6.0	2.5	1.2
10	45	Male	2.1	0.8	310	55	70	7.5	4.0	0.8
11	62	Female	1.8	0.7	220	40	35	6.8	3.1	1.0
12	50	Male	4.0	1.2	400	60	80	7.2	3.5	1.1
13	50	Female	1.0	0.3	191	22	31	7.8	4.0	1.0
14	52	Female	0.6	0.1	194	10	12	6.9	3.3	0.9
15	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
16	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0

B.3 Audit Trace: Fold 3 | Ctx: 40% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	36	Male	0.8	0.2	158	29	39	6.0	2.2	0.5
2	4	Male	0.8	0.2	460	152	231	6.5	3.2	0.9
3	32	Male	3.7	1.6	612	50	88	6.2	1.9	0.4
4	36	Male	5.3	2.3	145	32	92	5.1	2.6	1.0
5	65	Male	1.1	0.3	258	48	40	7.0	3.9	1.2
6	39	Male	0.6	0.2	188	28	43	8.1	3.3	0.6
7	29	Female	1.2	0.4	210	30	25	6.5	3.2	0.8
8	45	Male	0.8	0.3	150	40	35	7.0	3.5	0.9
9	38	Female	2.5	1.0	180	50	45	7.2	3.8	1.0
10	52	Male	0.9	0.2	250	60	55	6.8	3.0	0.9
11	34	Female	1.5	0.6	220	35	30	6.3	2.9	0.7
12	41	Male	0.7	0.2	190	45	50	7.1	3.4	1.1
13	44	Female	1.9	0.6	298	378	602	6.6	3.3	1.0
14	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
15	53	Male	0.9	0.4	238	17	14	6.6	2.9	0.8
16	28	Male	0.8	0.3	190	20	14	4.1	2.4	1.4

B.4 Audit Trace: Fold 4 | Ctx: 40% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	28	Female	0.6	0.1	137	22	16	4.9	1.9	0.6
2	22	Male	0.8	0.2	300	57	40	7.9	3.8	0.9
3	32	Male	0.7	0.2	165	31	29	6.1	3.0	0.96
4	50	Male	0.6	0.2	137	15	16	4.8	2.6	1.1
5	27	Male	1.2	0.4	179	63	39	6.1	3.3	1.1
6	40	Male	0.7	0.1	202	37	29	5.0	2.6	1.0
7	29	Female	0.5	0.1	150	30	25	6.0	3.2	0.87
8	34	Male	1.0	0.3	200	45	35	7.0	3.8	1.0
9	41	Female	0.8	0.2	180	40	30	6.5	3.0	0.75
10	55	Male	2.5	1.0	300	70	60	8.0	4.0	1.2
11	38	Female	1.2	0.4	220	55	45	7.5	3.5	0.9
12	26	Male	0.9	0.2	190	35	28	6.8	3.3	1.1
13	42	Male	0.7	0.2	197	64	33	5.8	2.4	0.7
14	69	Male	0.9	0.2	215	32	24	6.9	3.0	0.7
15	53	Male	1.6	0.9	178	44	59	6.5	3.9	1.5
16	49	Male	1.3	0.4	206	30	25	6.0	3.1	1.06

B.5 Audit Trace: Fold 5 | Ctx: 40% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	75	Male	2.8	1.3	250	23	29	2.7	0.9	0.5
2	75	Male	10.6	5.0	562	37	29	5.1	1.8	0.5
3	17	Male	0.9	0.2	279	40	46	7.3	4.0	1.2
4	73	Male	1.8	0.9	220	20	43	6.5	3.0	0.8
5	26	Male	6.8	3.2	140	37	19	3.6	0.9	0.3
6	36	Female	0.7	0.2	152	21	25	5.9	3.1	1.1
7	29	Female	1.2	0.5	180	25	30	6.5	3.0	0.8
8	34	Male	0.9	0.3	210	40	50	7.0	3.5	1.0
9	45	Female	2.5	1.1	250	35	45	8.0	4.0	1.2
10	52	Male	3.0	1.5	300	50	60	7.5	3.8	1.1
11	67	Female	0.8	0.2	190	20	25	6.0	2.5	0.7
12	39	Male	1.5	0.7	220	45	55	7.8	4.2	1.0
13	33	Male	2.1	0.7	205	50	38	6.8	3.0	0.7
14	22	Female	6.7	3.2	850	154	248	6.2	2.8	0.8
15	48	Male	5.8	2.5	802	133	88	6.0	2.8	0.8
16	84	Female	0.7	0.2	188	13	21	6.0	3.2	1.1

B.6 Audit Trace: Fold 6 | Ctx: 40% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	60	Male	0.9	0.3	168	16	24	6.7	3.0	0.8
2	32	Male	0.7	0.2	165	31	29	6.1	3.0	0.96
3	48	Female	1.4	0.8	621	110	176	7.2	3.9	1.1
4	78	Male	1.0	0.3	152	28	70	6.3	3.1	0.9
5	34	Female	0.6	0.1	161	15	19	6.6	3.4	1.0
6	55	Male	75.0	3.6	332	40	66	6.2	2.5	0.6
7	29	Female	0.8	0.3	210	25	30	6.5	3.2	1.1
8	54	Male	1.2	0.5	300	40	50	7.0	3.5	0.9
9	37	Female	0.9	0.4	180	22	25	6.8	3.4	1.2
10	45	Male	2.1	1.0	250	35	40	7.2	3.6	1.0
11	62	Female	0.7	0.2	220	30	28	6.9	3.1	1.3
12	50	Male	1.5	0.6	290	45	55	7.5	3.8	0.8
13	50	Male	0.9	0.3	194	190	73	7.5	3.9	1.0
14	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
15	18	Male	0.8	0.2	228	55	54	6.9	4.0	1.3
16	30	Female	0.7	0.2	194	32	36	7.5	3.6	0.92

B.7 Audit Trace: Fold 7 | Ctx: 40% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	43	Male	22.5	11.8	143	22	143	6.6	2.1	0.46
2	42	Female	0.8	0.2	168	25	18	6.2	3.1	1.0
3	34	Female	0.8	0.2	192	15	12	8.6	4.7	1.2
4	41	Male	0.9	0.2	169	22	18	6.1	3.0	0.9
5	26	Female	0.6	0.2	142	12	32	5.7	2.4	0.75
6	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
7	29	Female	1.2	0.4	150	30	25	6.5	3.0	0.8
8	54	Male	0.9	0.3	200	50	40	7.2	3.5	0.9
9	37	Female	2.5	0.9	180	45	35	6.8	3.2	1.1
10	45	Male	1.1	0.5	250	70	60	7.5	3.8	0.7
11	62	Female	0.8	0.2	220	40	30	6.0	2.5	0.6
12	34	Male	1.6	0.7	190	55	45	7.0	3.1	1.0
13	13	Female	0.7	0.2	350	17	24	7.4	4.0	1.1
14	54	Female	23.2	12.6	574	43	47	7.2	3.5	0.9
15	63	Male	0.9	0.2	194	52	45	6.0	3.9	1.85
16	46	Male	0.6	0.2	290	26	21	6.0	3.0	1.0

B.8 Audit Trace: Fold 8 | Ctx: 40% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	60	Male	8.6	4.0	298	412	850	7.4	3.0	0.6
2	56	Male	1.0	0.3	195	22	28	5.8	2.6	0.8
3	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
4	60	Male	19.6	9.5	466	46	52	6.1	2.0	0.4
5	28	Female	0.9	0.2	316	25	23	8.5	5.5	1.8
6	52	Male	1.8	0.8	97	85	78	6.4	2.7	0.7
7	34	Female	1.5	0.6	210	35	40	6.5	3.2	0.8
8	45	Male	2.2	1.0	300	50	60	7.0	3.5	0.9
9	29	Female	0.9	0.3	180	25	30	6.0	2.8	1.1
10	52	Male	3.1	1.5	250	40	45	6.8	3.0	0.7
11	38	Male	1.8	0.9	220	30	35	7.2	3.4	1.0
12	60	Female	4.0	1.8	400	60	70	7.5	3.8	0.8
13	48	Male	0.7	0.2	208	15	30	4.6	2.1	0.8
14	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
15	27	Male	1.0	0.3	180	56	111	6.8	3.9	1.85
16	28	Female	0.8	0.2	309	55	23	6.8	4.1	1.51

B.9 Audit Trace: Fold 9 | Ctx: 40% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	40	Male	3.6	1.8	285	50	60	7.0	2.9	0.7
2	60	Male	0.5	0.1	500	20	34	5.9	1.6	0.37
3	50	Male	0.7	0.2	188	12	14	7.0	3.4	0.9
4	14	Male	0.9	0.3	310	21	16	8.1	4.2	1.0
5	18	Male	0.8	0.2	228	55	54	6.9	4.0	1.3
6	69	Female	0.8	0.2	146	42	70	8.4	4.9	1.4
7	34	Female	1.2	0.5	210	30	40	6.5	3.0	1.1
8	45	Male	2.0	0.8	300	50	60	7.0	3.5	1.2
9	29	Female	0.9	0.3	150	20	25	6.8	3.2	1.0
10	55	Male	3.5	1.5	400	70	80	7.5	4.0	1.3
11	60	Female	1.1	0.4	180	25	35	6.0	2.8	1.2
12	42	Male	2.5	1.0	250	40	50	6.7	3.1	1.1
13	26	Male	2.0	0.9	195	24	65	7.8	4.3	1.2
14	47	Male	2.7	1.3	275	123	73	6.2	3.3	1.1
15	74	Male	1.0	0.3	175	30	32	6.4	3.4	1.1
16	48	Male	3.2	1.6	257	33	116	5.7	2.2	0.62

B.10 Audit Trace: Fold 10 | Ctx: 40% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	24	Male	1.0	0.2	189	52	31	8.0	4.8	1.5
2	39	Male	3.8	1.5	298	102	630	7.1	3.3	0.8
3	38	Male	0.8	0.2	247	55	92	7.4	4.3	1.38
4	25	Female	0.7	0.1	140	32	25	7.6	4.3	1.3
5	75	Female	0.8	0.2	188	20	29	4.4	1.8	0.6
6	66	Female	4.2	2.1	159	15	30	7.1	2.2	0.4
7	29	Female	1.2	0.5	210	30	25	6.8	3.5	1.2
8	34	Male	0.8	0.2	175	40	30	7.0	3.7	1.5
9	50	Female	2.5	1.0	300	50	60	6.5	3.2	0.9
10	45	Male	3.0	1.5	250	70	80	7.5	4.0	1.1
11	60	Female	0.9	0.3	180	20	22	6.2	3.0	1.0
12	38	Male	2.1	0.8	220	35	28	6.9	3.6	1.3
13	54	Female	5.5	3.2	350	67	42	7.0	3.2	0.8
14	43	Male	22.5	11.8	143	22	143	6.6	2.1	0.46
15	57	Male	4.5	2.3	315	120	105	7.0	4.0	1.3
16	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3